Email Thread: CR-0001 Impact on HV Junction Box Requirements

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From: Katrin Jansen (katrin.jansen@nova-ev.example) **To:** Ines Brown, Ulla Becker **Cc:** Zain Kim
Date: 2026-06-01 09:12 **Subject:** CR-0001 impact on HV Junction Box requirements

Hi Ines, hi Ulla,

I am preparing the Change Control Board presentation for CR-0001 (Raise insulation resistance threshold across HVB components) on 2026-06-12. The change follows the new UN ECE R100 working summary, rule R100-6.1.1, threshold 500 Ω/V instead of 100 Ω/V.

Three of the nine affected requirements belong to the HV Junction Box (CMP-HVB-04): REQ-HVB-0003 (Detect insulation faults, owner Ines), REQ-HVB-0095 (Balance cell voltages, owner Ines) and REQ-HVB-0117 (Log cell temperatures, owner Ulla). Could you both confirm the impact on your requirements and on the test sets? My estimate for the whole change is 61,530.37 EUR, almost all of it retest.

Thanks, Katrin

From: Ines Brown (ines.brown@nova-ev.example) **To:** Katrin Jansen **Cc:** Ulla Becker, Zain Kim
Date: 2026-06-01 15:40 **Subject:** RE: CR-0001 impact on HV Junction Box requirements

Hi Katrin,

Confirmed for REQ-HVB-0003 and REQ-HVB-0095. REQ-HVB-0003 has two test sets, TS-HVB-0003 and TS-HVB-0004, both required for Design Validation Complete (MS-D). Both need a full re-execution because the threshold is checked in the "Boundary values" test case and the other cases depend on it. REQ-HVB-0095 shares the measurement path with the insulation monitoring, so its test set has to be re-executed as well, even though the balancing logic itself does not change.

The threshold is a calibration parameter per DG-HV-006, so no firmware release is needed. Please state that clearly in the CCB slides, it is the main reason the cost is that low.

Ines

From: Ulla Becker (ulla.becker@nova-ev.example) **To:** Katrin Jansen **Cc:** Ines Brown, Zain Kim
Date: 2026-06-02 08:05 **Subject:** RE: CR-0001 impact on HV Junction Box requirements

Katrin,

REQ-HVB-0117 (HV Junction Box: Log cell temperatures) is affected only because the temperature log also records the insulation value at each sample, and the plausibility check in the logger uses the threshold. One test set, one re-execution. I agree with the estimate.

One request: the retest for the Cell Module requirements REQ-HVB-0047, REQ-HVB-0081 and REQ-HVB-0144 and for the Battery Management System requirements REQ-HVB-0116, REQ-HVB-0127 and REQ-HVB-0025 should be planned together with ours, otherwise we book the HIL bench three times.

Ulla

From: Zain Kim (zain.kim@nova-ev.example) **To:** Katrin Jansen, Ines Brown, Ulla Becker **Cc:** David Kovacs **Date:** 2026-06-02 11:30 **Subject:** RE: CR-0001 impact on HV Junction Box requirements

All,

From the change management side: CR-0001 qualifies for CCP-04, regulatory changes are approved by default unless Functional Safety sees a risk. I have asked Kenji Wagner to confirm before the board. Because the change affects three components, CCP-10 applies and Katrin is the coordinating engineer.

David, can you reserve the HIL bench for a combined retest block after the board decision?

Zain

From: Katrin Jansen (katrin.jansen@nova-ev.example) **To:** Ines Brown, Ulla Becker, Zain Kim, David Kovacs **Date:** 2026-06-03 10:02 **Subject:** RE: CR-0001 impact on HV Junction Box requirements

Thanks all. Summary for the record:

Decision: The retest of all nine requirements affected by CR-0001 is executed as one combined block on the HIL bench, coordinated by me, with the HV Testers. Rationale: the nine test sets share the measurement path and the bench; three separate bookings would cost weeks. Based on CCP-10 and LL-08. Status: Approved by the four of us, to be confirmed by the Change Control Board on 2026-06-12. Affects CR-0001, REQ-HVB-0003, REQ-HVB-0095, REQ-HVB-0117, REQ-HVB-0047, REQ-HVB-0081, REQ-HVB-0144, REQ-HVB-0116, REQ-HVB-0127 and REQ-HVB-0025.

David has pencilled in the HIL bench from 2026-08-17. I will present this at the board.

Katrin